Project Proposal

AI-Powered Worker Tracking & Behavioral Analytics System

1.1 Project Overview

The AI-Powered Worker Tracking & Behavioral Analytics System is an intelligent video monitoring solution designed to improve security awareness and operational visibility in industrial environments. The system uses Computer Vision and Artificial Intelligence to detect and track people, monitor predefined zones, identify specific movement patterns, and generate alerts when configured security or operational rules are triggered.

The initial version will use a single camera and support recorded video processing, with the possibility of integrating a live camera stream. The system will provide a centralized dashboard for viewing tracking results, reviewing incidents, and analyzing collected data. AWS cloud services will support application deployment, evidence storage, monitoring, and database management.

1.2 Problem Statement

Traditional video surveillance systems require security personnel to continuously monitor camera feeds and manually identify potentially important events. This process can be time-consuming, inconsistent, and difficult to scale.

Furthermore, conventional surveillance systems may not provide structured tracking histories, automated zone monitoring, or searchable incident records. These limitations can delay incident detection and make it difficult to analyze movement patterns over time.

1.3 Project Objectives

- Automate person detection and tracking in video footage.
- Assign temporary tracking IDs to detected individuals and maintain their tracks over time.
- Monitor predefined restricted zones and virtual boundaries.
- Detect rule-based events such as unauthorized zone entry, excessive zone dwell time, and prolonged inactivity.
- Generate incident records with timestamps, event types, and supporting evidence.

- Develop a centralized dashboard for monitoring tracks, reviewing incidents, and viewing reports.
- Deploy the application using appropriate AWS services.
- Evaluate the system using measurable Computer Vision, performance, and reliability metrics.

1.4 Project Scope

In Scope

- Single-camera video processing.
- Person detection using a pretrained object detection model.
- Single-camera multi-object tracking.
- Configurable zones and virtual lines.
- Rule-based incident detection.
- Incident history and evidence management.
- Dashboard and REST API development.
- Database integration and AWS deployment.
- Model evaluation, functional testing, and cost monitoring.

Out of Scope

- Cross-camera identity matching in the initial version.
- Facial recognition or biometric identification.
- Guaranteed identification of individual workers without a reliable verification process.
- Automatic conclusions about employee intent, misconduct, or productivity.
- Theft detection without a clearly defined and validated detection scenario.
- Fully autonomous employment or disciplinary decisions.
- Guaranteed performance under all lighting, occlusion, and camera conditions.

1.5 Expected Deliverables

- A working Computer Vision detection and tracking pipeline.
- A backend API and database.
- A web-based monitoring dashboard.
- An incident detection and evidence storage module.
- An AWS-hosted application.
- A testing report documenting accuracy, limitations, and performance.
- Technical documentation and a final project presentation.

1.6 Target Users

- Factory security personnel.
- Operations supervisors.
- Authorized system administrators.